Function Operations

Name _____

Algebra 2 Lesson 5-5 (fast track) Date _____ Period _____

Learning goal: I can write the rule and state the domain of a composite function, and use the order of a composition to answer a real-world question.

Language goal: I can explain how I composed two functions using the words composition, inside function, substitute, order and domain.

On the quiz: (1) $g(f(x)) + g(x)$: *substitute* the whole rule for f into g (Example 5B, Try It 5), then add $g(x)$; the wrong *order*, $f(g(x))$, is a choice. (2) Revenue and (3) a quotient $\frac{f}{g}$ and its *domain* (full Savvas quiz only). (4, bonus) Work inside out with square roots: $k(3)$ first, then h of that number (Example 5A, the Exit question). (5) The rule and *domain* of $f \circ g$ when the inside function g has the square root (Example 5A, checks 1 and 2).

On the Topic 5 test: Question 11 asks for the domain of $f \circ g$ with $f(x) = \sqrt{x}$ (Example 5). Questions 19 and 21 ask which composition models a sale price or a sphere's volume (Example 6).

Composite function. Apply the rule for g, then apply the rule for f to the result:

$(f \circ g)(x) = f(g(x))$ g is the **inside function** (first); f is the outside function (second).

The operation $\circ$ is called **composition of functions**. The **domain** of $f \circ g$ is every x in the domain of g (check 1) for which $g(x)$ is in the domain of f (check 2).
Order matters: for $f(x) = 3x + 5$, $g(x) = x - 3$: $f \circ g = 3(x - 3) + 5 = 3x - 4$,
but $g \circ f = (3x + 5) - 3 = 3x + 2$.

Example 5: Write a Rule for a Composite Function

What is the rule for the composition $(f \circ g)(x)$? *Substitute* the rule for g wherever x appears in f.

A. $f(x) = \sqrt{x + 7}$ and $g(x) = 2x - 5$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(2x - 5) \\ &= \sqrt{(2x - 5) + 7} \\ &= \sqrt{2x + 2}\end{aligned}$$

Domain. g takes all real numbers; under the root, $2x + 2 \geq 0$, so the domain of $f \circ g$ is $x \geq -1$.
Check $x = 1$: $g(1) = -3$, $f(-3) = \sqrt{4} = 2$; and $\sqrt{2(1) + 2} = 2$. ✓

B. $f(x) = x^2 + x + 2$ and $g(x) = 4 - x$

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(4 - x) \\ &= (4 - x)^2 + (4 - x) + 2 \\ &= 16 - 8x + x^2 + 4 - x + 2 \\ &= x^2 - 9x + 22\end{aligned}$$

Domain. All real numbers.
Watch: *every* x in f is replaced, including the lone x: it becomes $(4 - x)$.
Domain checks: in A, check 2 sets the limit; when g has the square root, check 1 does.

Say it: To find $(f \circ g)(x)$ in Part A, I _____ the inside function _____ for x in f. The domain is $x \geq -1$ because ...

Try It 5: Write the rules for two compositions

Identify the rules for $f \circ g$ and $g \circ f$.

Hint: Write $f(g(x))$ and $g(f(x))$ first, then substitute and simplify.

a. $f(x) = x^3,\ \ g(x) = x + 1$

$(f \circ g)(x) = f(g(x)) =$

b. $f(x) = 3\sqrt{2x},\ \ g(x) = 4x + 11$

$(f \circ g)(x) = f(g(x)) =$

$(g \circ f)(x) = g(f(x)) =$

$(g \circ f)(x) = g(f(x)) =$

Say it: In part a, $f \circ g$ and $g \circ f$ are _____ (the same / different), so the _____ of a composition _____ (matters / does not matter), because ...

Example 6: Use a Composite Function Model

A clothing store posts two discounts on social media. Followers may use both; they tell the cashier in which order to apply them. In which order should they ask for the discounts to save the most money?

EXCLUSIVELY FOR OUR SOCIAL MEDIA FOLLOWERS! **$5 OFF** NEXT PURCHASE	EXCLUSIVELY FOR OUR SOCIAL MEDIA FOLLOWERS! **10% OFF** NEXT PURCHASE

Let x be the price of the purchase. Model each discount with a function:
$f(x) = x - 5$ ($5 off) $g(x) = x - 0.1x = 0.9x$ (price minus 10% of the price)

10% off, then $5 off: the rule for $f \circ g$ (g first).

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(0.9x) \\ &= 0.9x - 5\end{aligned}$$

$5 off, then 10% off: the rule for $g \circ f$ (f first).

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(x - 5) \\ &= 0.9(x - 5) = 0.9x - 4.5\end{aligned}$$

Test a price. A shirt costs $30.00 before the discounts.
$(f \circ g)(30) = 0.9(30) - 5 = 22$, or $22.00. $(g \circ f)(30) = 0.9(30) - 4.5 = 22.5$, or $22.50.

Answer: ask for **10% off first, then $5 off**. The 10% is taken off the larger price, so it saves more.

Say it: The order "10% first, then $5" is the composition _____, because g is the _____. It is the better deal because ...

Try It 6: Model a discount and a sales tax

As a member of the Games Shop rewards program, you get a 20% discount on purchases. All sales are subject to a 6% sales tax. Write functions to model the discount and the sales tax, then identify the rule for the composition function that calculates the final price you would pay at Games Shop.

Let $x =$ _____

Discount: $d(x) =$ _____

Sales tax: $t(x) =$ _____

Final price (which function is the inside function?):

(____ $\circ$ ____)$(x) =$ _____

Say it: In my composition the inside function is _____, so the _____ is applied first. Switching the order here _____ (changes / does not change) the final price, because ...

IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
EKJ ★	Composition of linear and quadratic functions: find an equation	Quiz 1: $g(f(x))$ with a quadratic inside; Example 5B, Try It 5a
RSP ★	Composition of linear functions: find an equation	Quiz 1 and 5: the rule for $f \circ g$ in the right order; Example 6, Try It 6
P9T ★	Composition of linear and quadratic functions: find a value	Quiz 4 (bonus): inside out, one number at a time
QQD ★	Add and subtract functions	Quiz 1: the $+\,g(x)$, by combining like terms
PS2	Evaluate functions	Warm-up, optional: substitute a number or an expression

Exit question: Compose two functions given as ordered pairs

Suppose two functions, $f(x)$ and $g(x)$, are only defined by the ordered pairs listed below.

$$f = (6,7),\ (5,2),\ (4,1),\ (10,8) \qquad g = (5,4),\ (3,6),\ (1,5),\ (2,10)$$

Find the ordered pairs that comprise $(f \circ g)(x)$.

x	$g(x)$ (inside)	$f(g(x))$ (outside)

$(f \circ g)(x) = \{$ _____ $\}$

Say it: The domain of $f \circ g$ is {_____} because each output of the inside function g is ...

_____

_____

_____

Before the quiz, finish these sentences.

To find $(f \circ g)(x)$, I substitute the inside function _____ for x in _____.

The domain of $f \circ g$ can be smaller than the domain of g because _____.

In a composition the order matters: $f \circ g$ applies _____ first, and $g \circ f$ applies _____ first.

Extension (self-paced, DOK 3, Practice 32): A music store takes \$5 off a purchase of \$20 or more, and you save an additional 15% off your total purchase when you open a charge account. Use composition of functions to find the sale price of a \$90 purchase when the \$5 off is applied first, then when the 15% off is applied first. In which order is the deal better for the consumer? Explain.